

Program : Diploma in Chemical Engineering	
Course Code : 3075	Course Title: Safety in Process Industries.
Semester : 3	Credits: No Credits
Course Category: Program Core	
Periods per week: 4 (L: 2 T: 2 P: 0)	Periods per semester: 60

Course Objectives:

- To impart thorough knowledge on the safe operations in chemical plants.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Unit operations and processes in chemical industries.		Chemical Process Principles	2

Course Outcomes :

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the special hazards of chemicals.	15	Understanding
CO2	Summarise the hazard identification techniques and personal protective equipments used in process plants.	18	Understanding
CO3	Identify the fire protection systems in a process plant.	15	Understanding
CO4	Summarise safe work permit system in process plants.	12	Understanding

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					2		
CO2					3		
CO3					2		
CO4					3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the special hazards of chemicals.		
M1.01	Summarise types of Hazards in chemical Industry	2	Understanding
M1.02	Summarise special hazards of chemicals.	4	Understanding
M1.03	Summarise the methods for Hazard survey.	4	Understanding
M1.04	Identify the reasons behind major industrial disasters.	5	Understanding
Contents: Types of hazard - Chemical Hazard - Physical Hazard - Ergonomic Hazard - Biological Hazard -Safety Hazard - Psychological Hazard –Risk - Differentiate Hazard and Risk. Toxicity – Flammability – Threshold Limit Value (TLV) or Permissible Exposure Limit (PEL) Hazard Survey – Inventory analysis, Dow Fire and Explosion Index, Mond Fire, Explosion and Toxicity Index. Summarise the industrial disasters and identify the reasons – Flixborough – Bhopal - Vishakhapattanam gas leak.			
CO2	Summarise the hazard identification techniques and personal protective equipments used in process plants.		
M2.01	Summarise the hazard identification techniques commonly used in process plants.	4	Understanding
M2.02	Identify the personal protective equipments for head, foot and hands.	6	Understanding
M2.03	Identify the personal protective equipments for whole body.	6	Understanding
M2.04	Identify the personal protective equipments for eye.	2	Understanding.
	Series Test – I		

Contents:

Summarise: Hazard and Operability Study - Preliminary Hazard Analysis - What if Analysis - Fault tree Analysis - Event Tree Analysis - Failure Modes and Effects Analysis

Head – Helmet, Foot – safety shoes, Hand – Nitrile, PVC coated fabric gloves, Leather Gloves, Cotton/Fabric Gloves

Whole body – PVC suits boiler suits, aprons, chemical suits

Eyes – face shield, goggles

Case studies- accidents occurred while doing duty without proper protective devices.

CO3	Identify the fire protection systems in a process plant.		
M3.01	Summarise the chemistry of fire.	2	Understanding
M3.02	Illustrate different detectors used in process plants.	5	Understanding
M3.03	Illustrate portable fire extinguishers.	3	Understanding
M3.04	Illustrate fixed fire protection system in process plants	5	Understanding

Contents:

Chemistry of fire - elements of fire and fire triangle - slow combustion - rapid combustion - spontaneous combustion - classify fire according to combustible materials - causes of initiation of fire - causes of electrical fire.

Criteria for selection of fire detectors - types of detectors in chemical plants - smoke detector - ionisation detector - photoelectric detector - infrared detector – fire alarm systems – LEL detector – H₂S detector.

Different types of portable fire extinguishers

Fire water storage – fire water pumps, electrical driven, diesel driven and jockey pump – fire water mains, pressure requirement – fire hydrants – fire water monitors – fire hose cabinets – deluge system – sprinkler system.

CO4	Summarise safe work permit system in process plants.		
M4.01	Summarise the significance of work permit system in process plants.	2	Understanding
M4.02	Summarise the permits for maintenance of equipments	3	Understanding
M4.03	Summarise work permit system	4	Understanding
M4.04	Summarise the checklists used in process plants.	3	Understanding
	Series Test – II		

Contents:

Need of permit system in process plants

Permit system for the maintenance of equipments – electrical lockout

Cold work permit - hot work permit- entry in to confined space- working under heights.

Checklists for startup and shutdown - hazard checklists

Text / Reference

T/R	Book Title/Author
T1	Introduction to chemical engineering; Waklter.L.Badger & Julius.T.Banchero
R1	Unit Operations; Warren.L.McCabe & Julian.C.Smith
R2	Unit Operations of Chemical engineering; P.Chathopadhyay
R3	Mechanical operations; Anup Kumar Swain, hemalatha patra and Gopendra Kishore Roy
R4	Chemical Engineers Hand Book; Robert H Perry.

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/107/103107156/